Hardware Support for Memory Safety

Mengjia Yan

Spring 2026

Overview

- What attacker can do with software bugs?
  - Demos, variations, real-world examples
- Hardware mitigations: what are the design tradeoffs?

Szekeres et al. SoK: Eternal War in Memory. SP'13

The Problem: Software Bugs

- Low-level Language Basics (C/C++/Assembly)
  - + Efficient, programmers have more control
  - - Bugs
  - - Programming productivity

- Widely used in production systems and legacy systems
  - Operating systems, web browsers, etc.
  - Large CVE numbers every year

The Problems of Using Pointers

- Pointer = Address of variables:
  - An 64-bit integer to indicate memory location where variable is stored
- It is programmers' responsibility to do **pointer check**, e.g. NULL, out-of-bound, use-after-free

Buffer Overflow Example

```
char * buffer1 = (char*)malloc(buffer1_size);
char * buffer2 = (char*)malloc(buffer2_size);

// fill up buffer2 with ‘B’
for (int i=0; i<buffer2_size; i++){
   buffer2[i] = 'B';
 }

// fill up buffer1 with ‘A’
for (int i=0; i< 2 * buffer1_size ; i++){
   buffer1[i] = 'A';
 }
```

Use-after-free Example

```
char * buffer1 = (char*)malloc(buffer1_size);
char * buffer2 = (char*)malloc(buffer2_size);
// fill buffer1 and buffer2 with ‘A’ and ‘B’ respectively

free(buffer1);
char *buffer3 = (char*)malloc(buffer3_size);
for (int i=0; i<buffer3_size; i++){
  buffer3[i] = 'C';
}
// print buffer 1 2 3
```

Memory Corruption Vulnerabilities

- Spatial safety:
  - out-of-bound (inter-object, intra-object)
  - Can happen on heap and stack

- Temporal safety:
  - Use-after-free
  - Use before initialization

- Why Python (and other high-level programming language) does not have these problems?
  - out-of-bound access => emit runtime checks
  - use-after-free => garbage collection

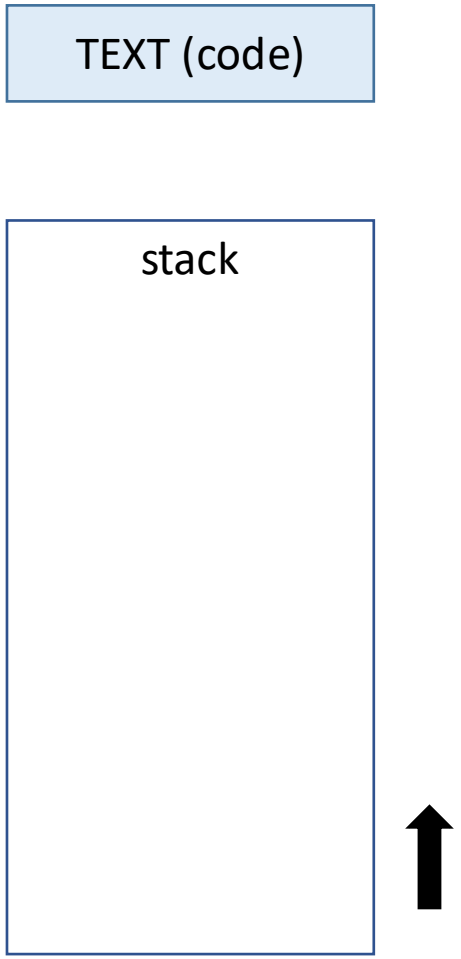
From Bugs to Exploitations

Via manipulating function pointers

Recap Stack Operations

```
int hello(){
    int x = 100;
    return x;
}

int main(){
    int a;
    int *p = &a;
    int b = hello();
ra  b = b+3;
    return 0;
}
```



Stack Smash: Manipulate the return address (ra)

```
// victim code
int func (char *str) {
  char buffer[12];
  strncpy(buffer, str, len(str));
  return 1;
}

int main() {
  …
  func (input);
  …
}
```

ra

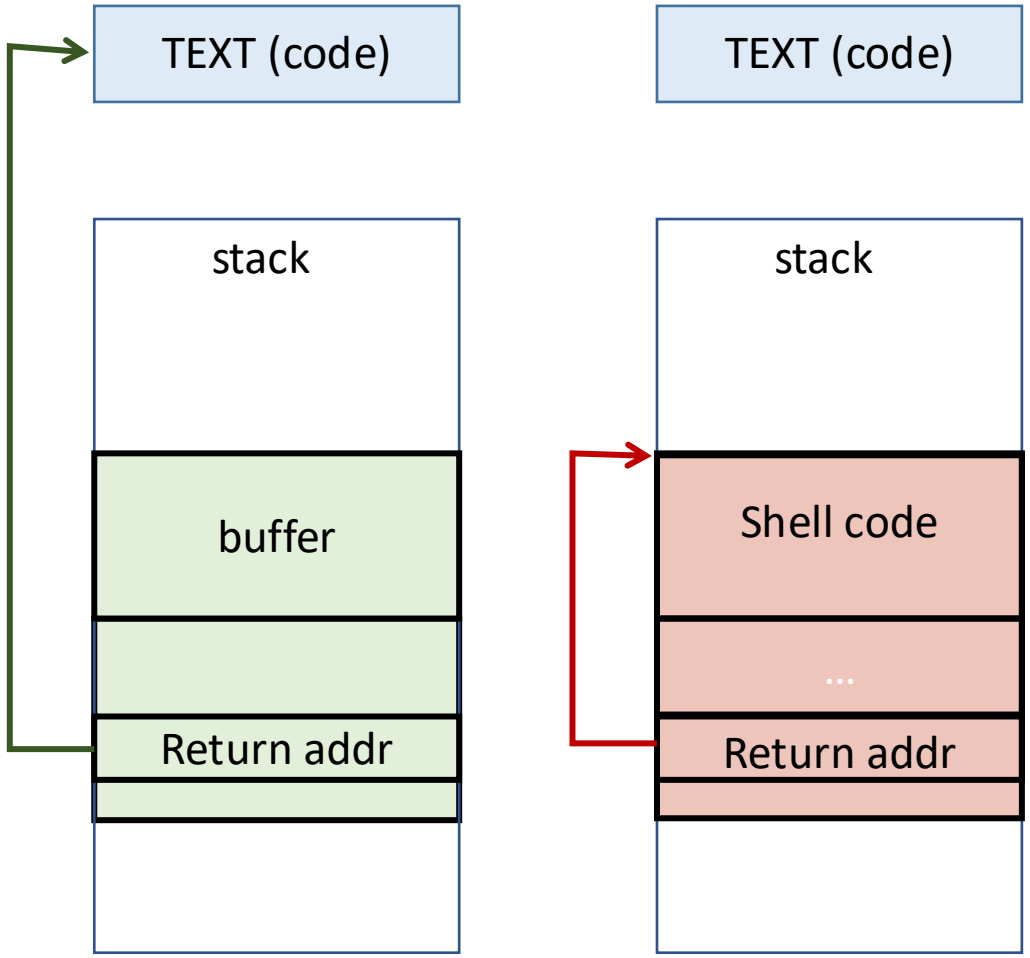
Shell code:

```
PUSH "/bin/sh"
CALL system
```

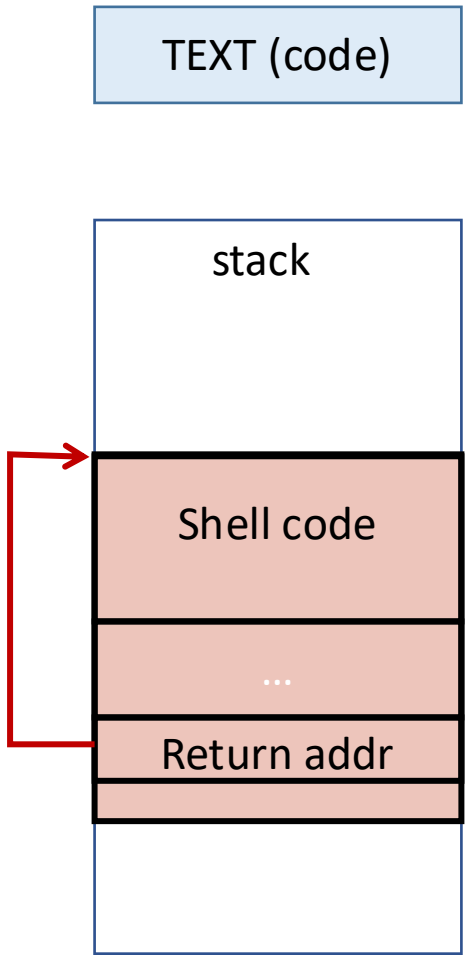
Input str:

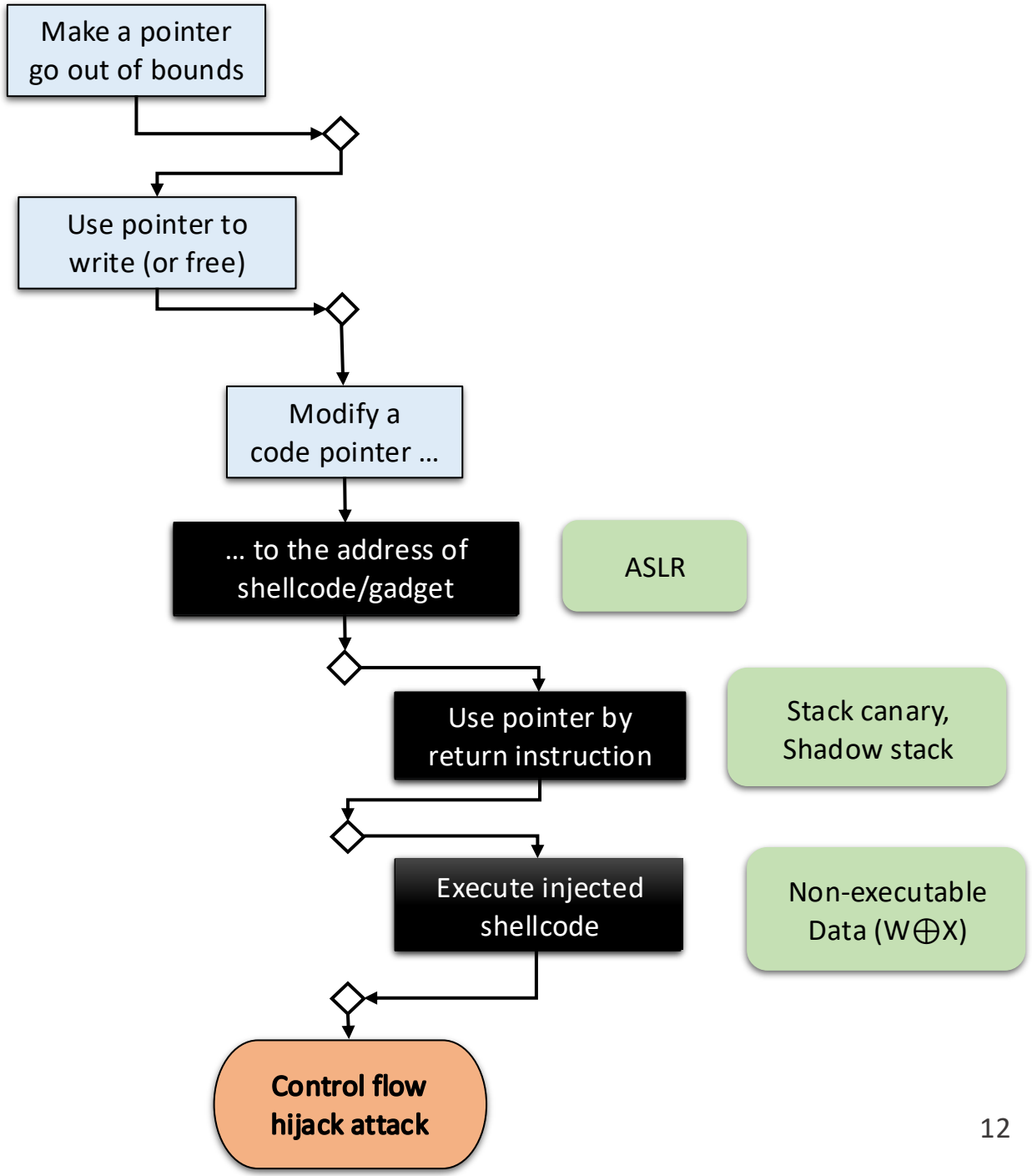
```
Shell code
.. Some padding..
Address of buffer
```

Stack Smash / Code Injection Attack

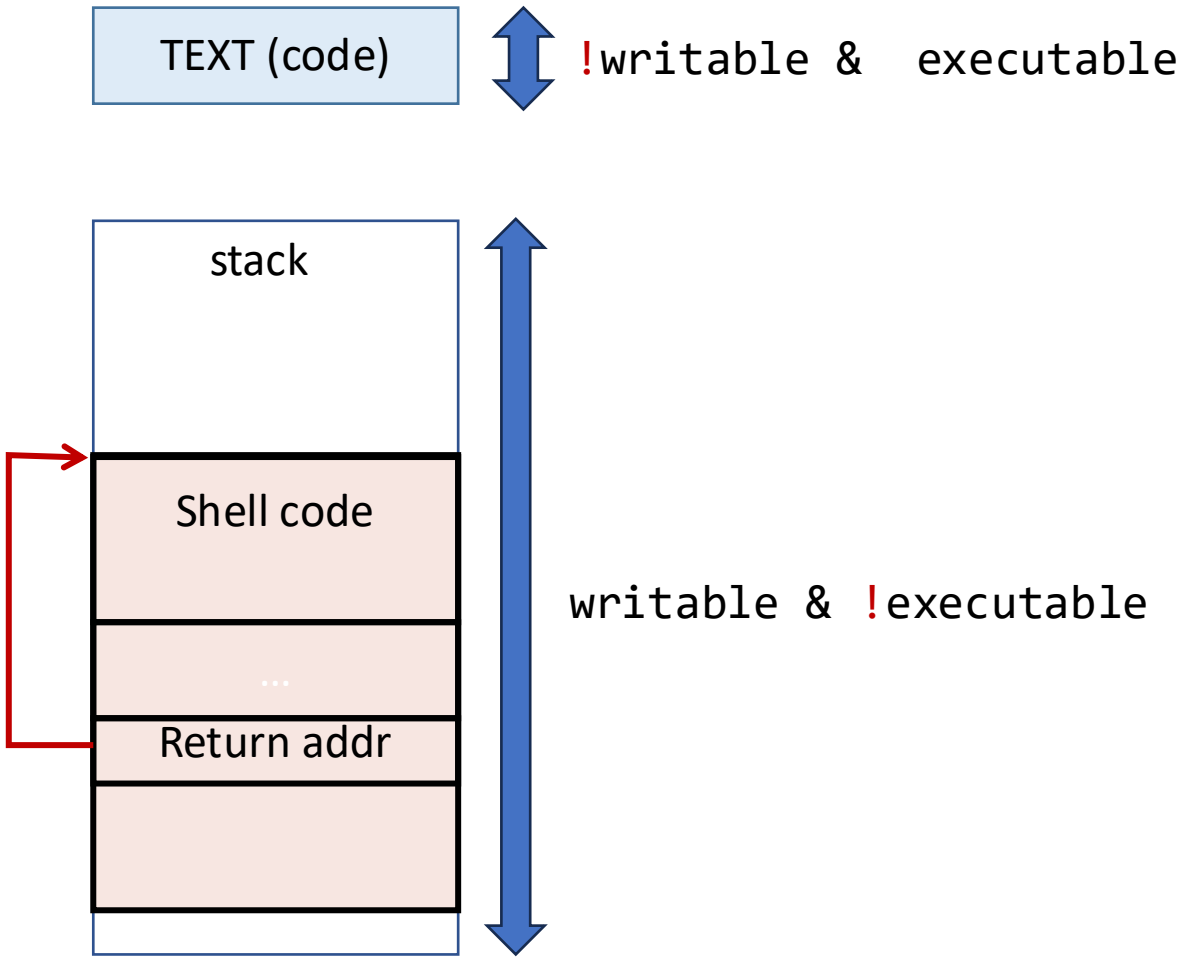


Mitigations



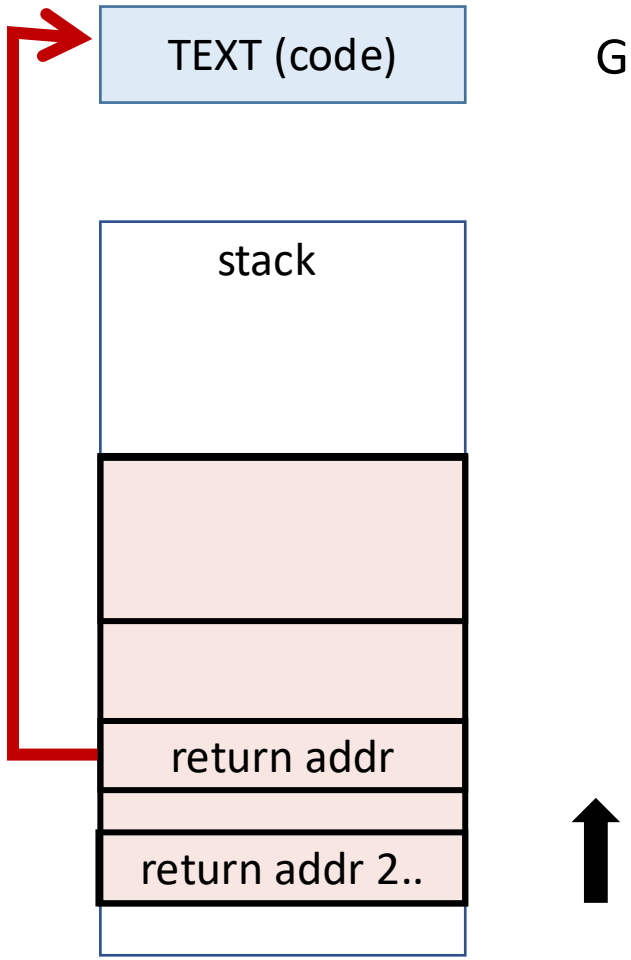


Return-Oriented Programming (ROP)



Stack Smash:
execute **injected** code

Blocked by "W⊕X"



ROP:
execute **pre-existing** code

Gadget example:

```
pop rdi
ret
```

An Attack Puzzle

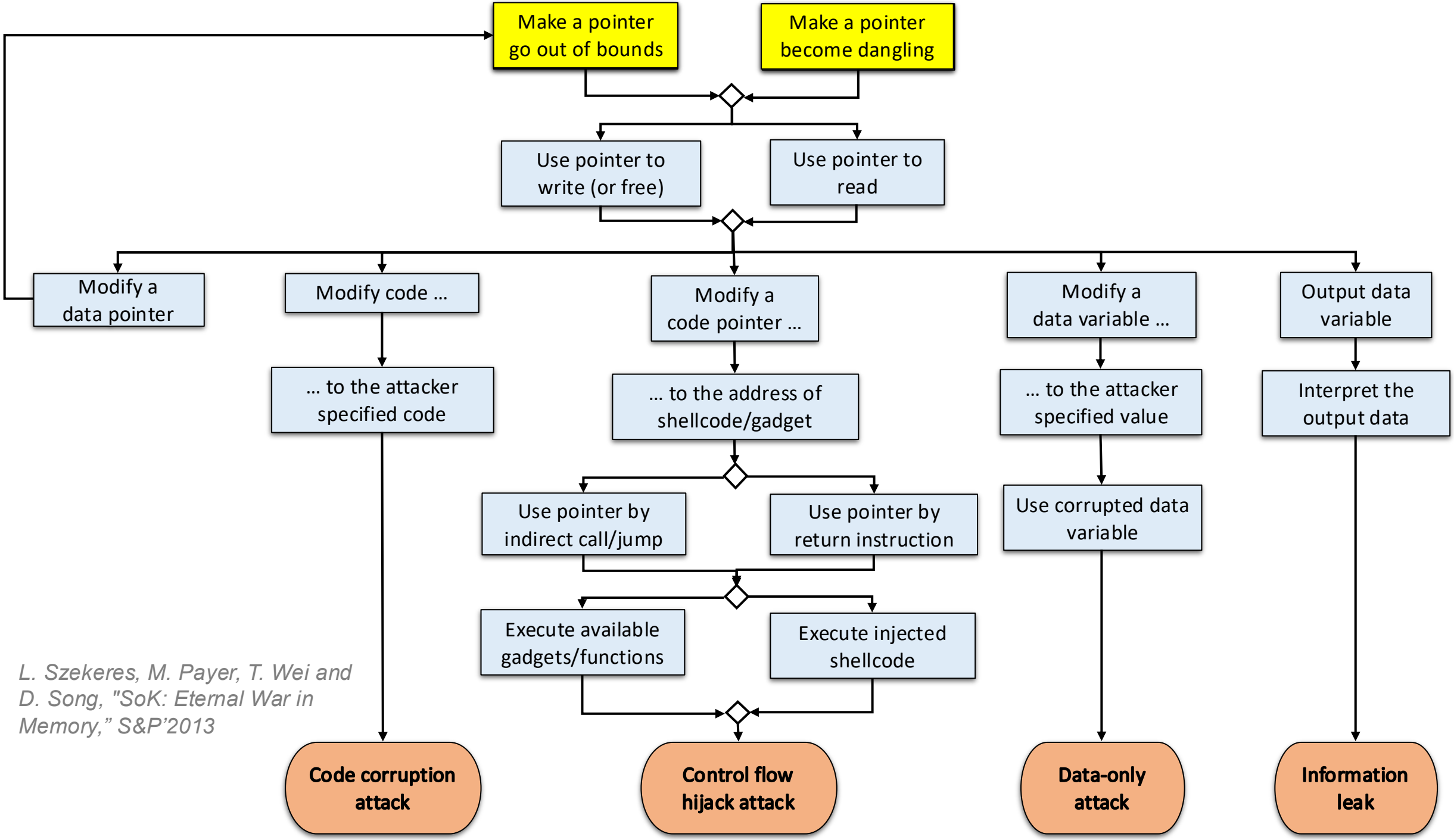
Jump-oriented programming

```
0x4011b6: int func0(u_int64_t s);
……
0x4011f2: int func1(u_int64_t s);
……
0x40122e: int hack(u_int64_t s);
……
```

```
int main(int argc, char *argv[]){
  // an array with two function pointers
  // to emulate a vtable
  int (*func_ptr[2])(u_int64_t) = {&func0, &func1};
  u_int64_t val;
  int choice;

  // obtain args from input; attacker-controlled
  choice = atoi(argv[1]);
  val = strtol(argv[2], NULL, 16);
  // index into the array to make a function call
  (*func_ptr[choice])(val);
}
```

Hardware Supported Mitigations



L. Szekeres, M. Payer, T. Wei and D. Song, "SoK: Eternal War in Memory," S&P'2013

Memory Safety

- Strongest security property that tries to address the problem at the root.

- Idea: include metadata and perform security checks at runtime
  - Spatial safety (bound information)
  - Temporal safety (allocation/de-allocation information)

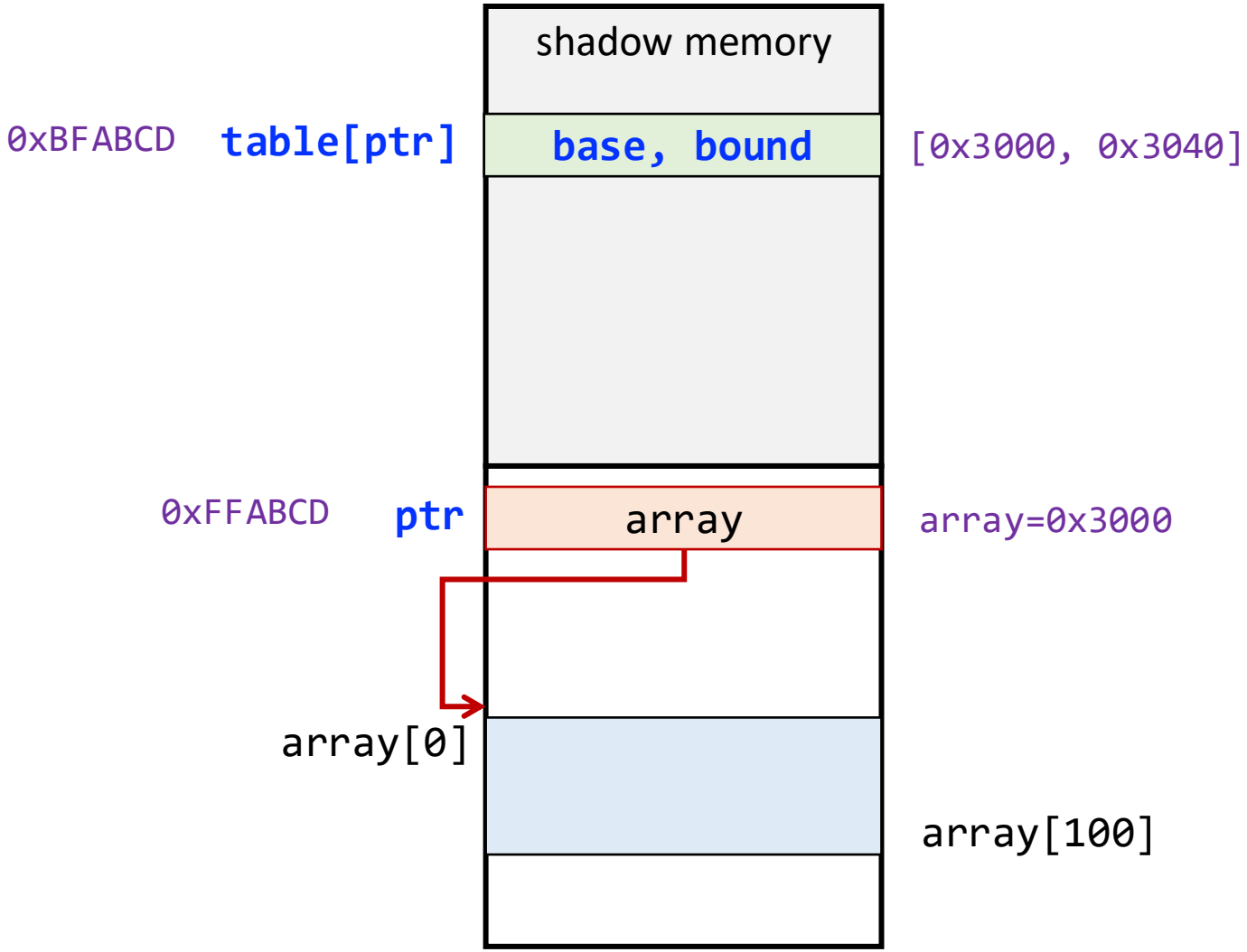
- Software solutions
  - Problem #1: performance overhead, extra instructions to perform the check
  - Problem #2: where to store metadata? -> in shadow memory

SoftBound: Highly Compatible and Complete Spatial Memory Safety for C; Nagarakatte et al; PLDI'09

SoftBound

Creating a pointer:

```
int array[100];
ptr = &array;
ptr_base = &array[0];
ptr_bound = &array[100];
table[ptr]={base, bound};
```



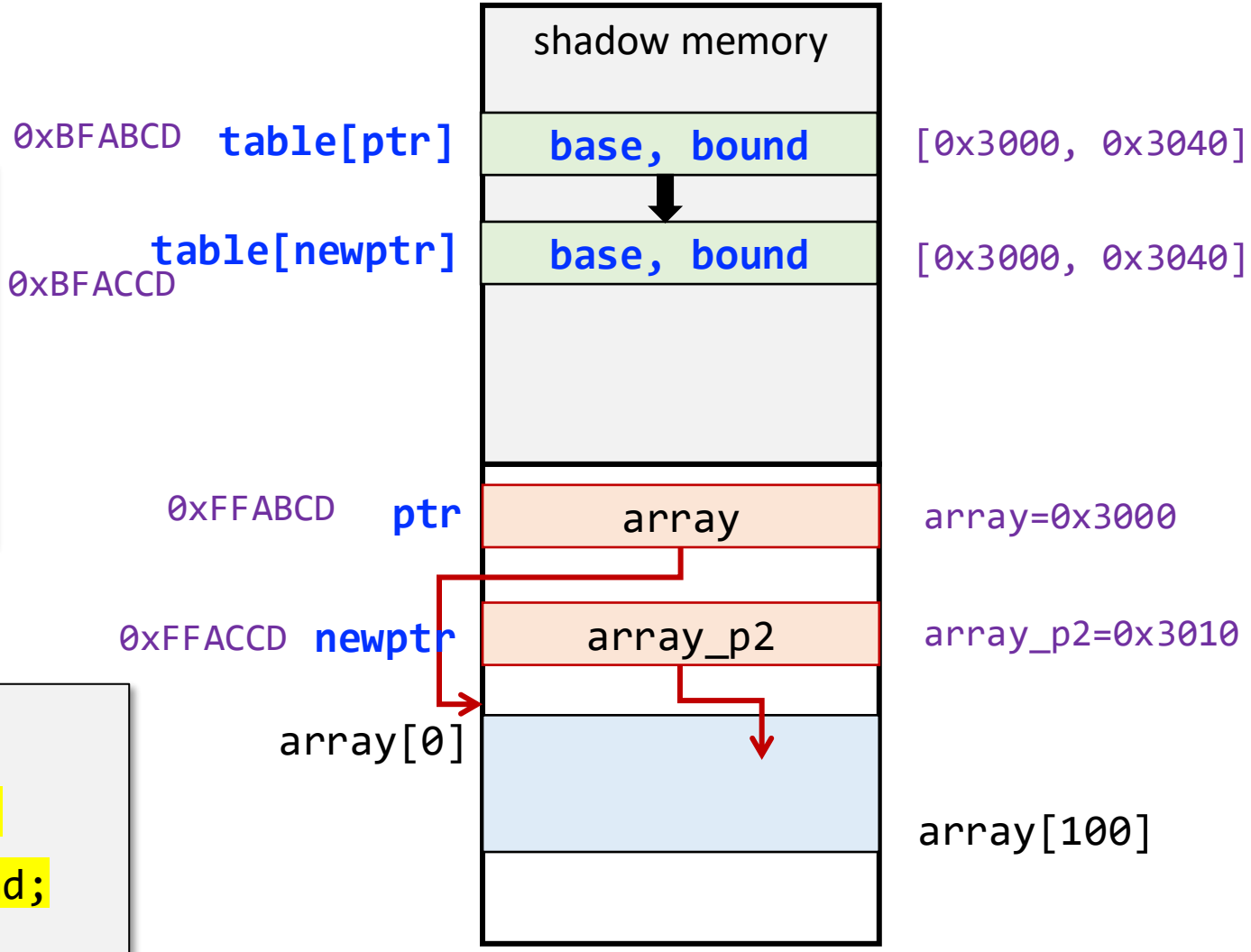
SoftBound

Creating a pointer:

```
int array[100];
ptr = &array;
ptr_base = &array[0];
ptr_bound = &array[100];
table[ptr]={base, bound};
```

Pointer arithmetic:

```
int* array_p2 = &array[10];
newptr_base = table[ptr].base;
newptr_bound = table[ptr].bound;
table[newptr]={base, bound};
```



SoftBound

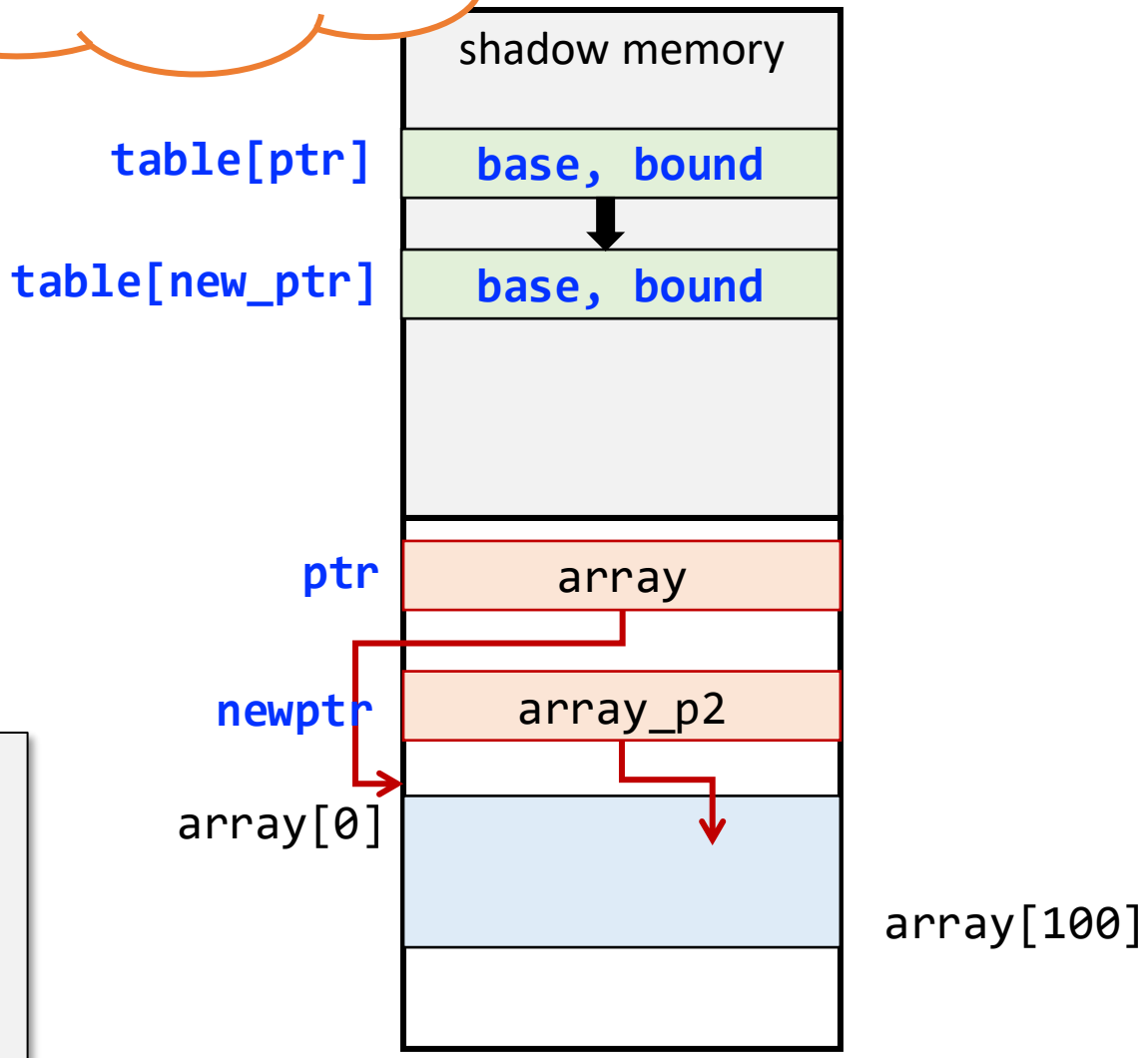
How many memory accesses do we need to do one array access?

Creating a pointer:

```
int array[100];
ptr = &array;
ptr_base = &array[0];
ptr_bound = &array[100];
table[ptr]={base, bound};
```

Check a pointer:

```
newptr = &array_p2;
{base, bound} = table[newptr];
if (base > array_p2 || bound …)
    go to err;
int* array_p2 = 0xFF;
```



HW Support for Memory Safety

A lot of work. The key is to understand the design trade-offs.

	Intel MPX (Memory Protection Extension)	**ARM MTE (Memory Tagging Extension)**
History	Announced in 2013, produced in 2015, now not supported anymore.	Introduced in ARM-8.5 in 2018. In 2019, Google announced that it is adopting Arm's MTE in Android. Apple will ship it soon.
Security		
Performance		
Compatibility		

Intel MPX (Memory Protection Extension)

4 bound registers (bnd0-3)

- `Bndmk`: create base and bound metadata
- `Bndldx/bndstx`: load/store metadata from/to bound tables
- `Bndcl/bndcu`: check pointer with lower and upper bounds

Dedicated hardware organized as tables to store metadata

```
int array[100];
ptr = &array;
ptr_base = &array[0];
ptr_bound = &array[100];
table[ptr]={base, bound};
```

```
int array[100];
ptr = &array;
bnd0 = bndmk(ptr, 100)

bndstx(&ptr,ptr,bnd0)
```

Intel MPX Explained: A Cross-layer Analysis of the Intel MPX System Stack; OLEKSENKO et al; SIGMETRICS'18

Intel MPX (Memory Protection Extension)

4 bound registers (bnd0-3)

- `Bndmk`: create base and bound metadata
- `Bndldx/bndstx`: load/store metadata from/to bound tables
- `Bndcl/bndcu`: check pointer with lower and upper bounds

Dedicated hardware organized as tables to store metadata

Pros and cons?

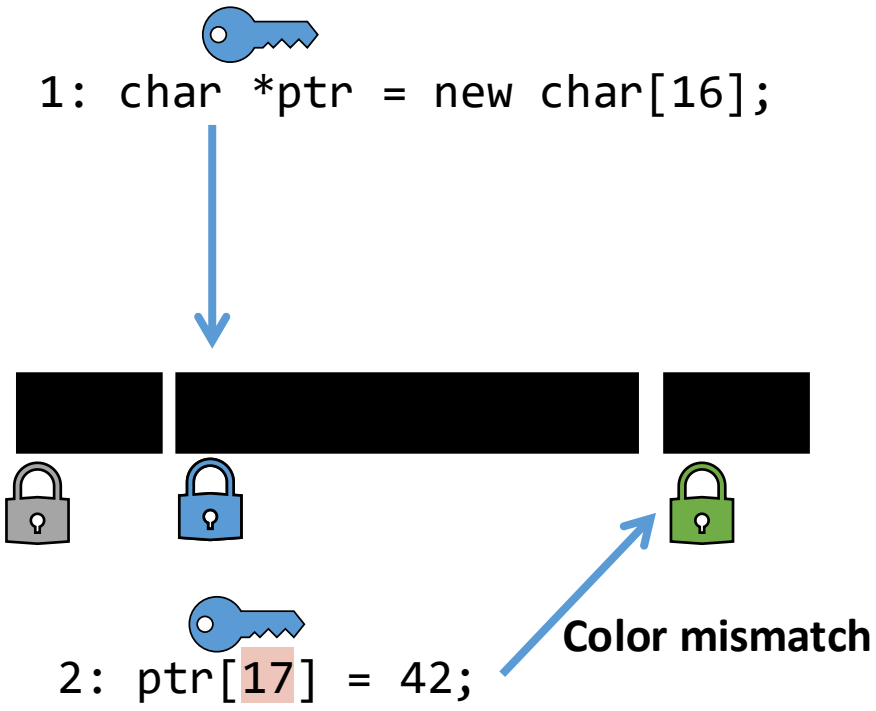
```
newptr = &array_p2;
{base, bound} = table[newptr];
if (base > array_p2 || bound …)
    go to err;
int* array_p2 = 0xFF;
```

→

```
newptr = &array_p2;
bnd1 = bndldx(&newptr, newptr)
bndcu(&newptr, bnd1)
bndcl(&newptr, bnd1)
int* array_p2 = 0xFF;
```

Intel MPX Explained: A Cross-layer Analysis of the Intel MPX System Stack; OLEKSENKO et al; SIGMETRICS'18

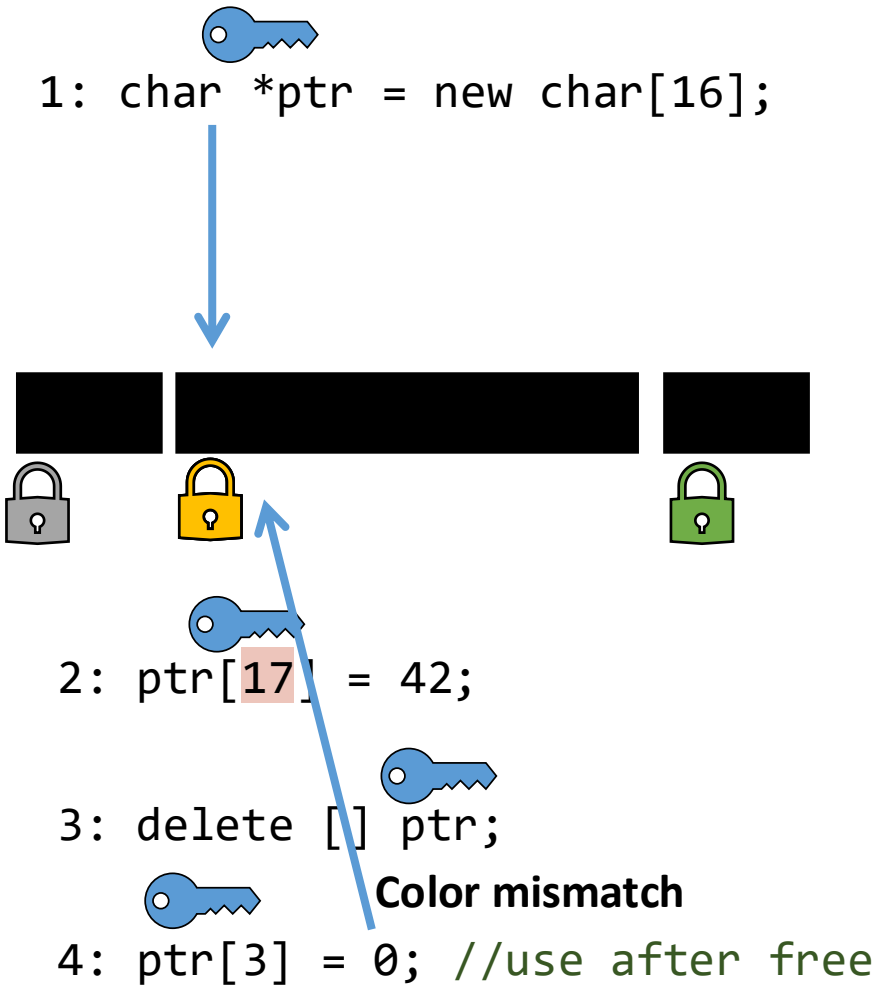
ARM MTE (Memory Tagging Extension)



- Memory locations are tagged by adding 4 bits of metadata (16 different **colors**)
- Each pointer also comes with 4 bit metadata (**keys**)
- Help with both spatial and temporal memory safety

Armv8.5-A Memory Tagging Extension White paper
https://security.googleblog.com/2019/08/adopting-arm-memory-tagging-extension.html

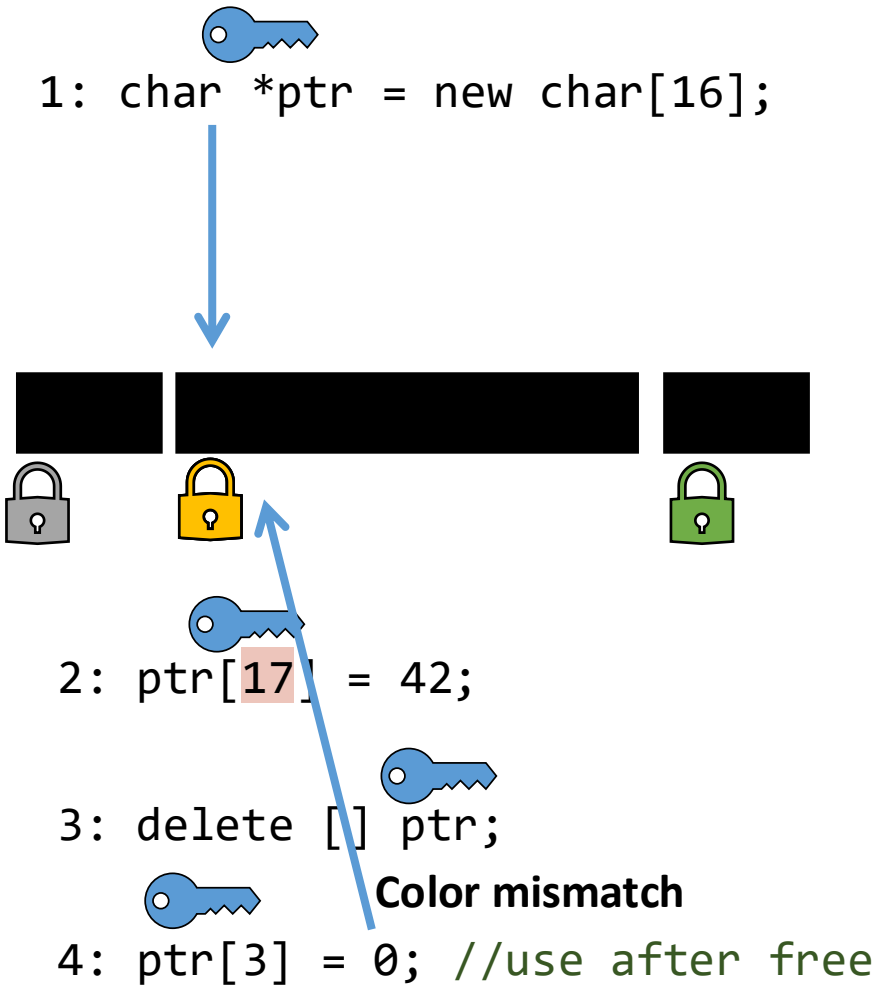
ARM MTE (Memory Tagging Extension)



- Memory locations are tagged by adding 4 bits of metadata (16 different **colors**)
- Each pointer also comes with 4 bit metadata (**keys**)
- Help with both spatial and temporal memory safety

Armv8.5-A Memory Tagging Extension White paper
https://security.googleblog.com/2019/08/adopting-arm-memory-tagging-extension.html

Analysis of ARM MTE



- Where to store tags (key and lock)?
  - Pointer tag is stored in top unused bits inside the pointer (no extra register needed)
  - Physical memory tag is stored in hardware (new hardware needed for both DRAM and cache)
- Limited tag bits
  - Cannot ensure two allocations have different colors
  - But can ensure that the tags of sequential allocations are always different

Armv8.5-A Memory Tagging Extension White paper
https://security.googleblog.com/2019/08/adopting-arm-memory-tagging-extension.html

HW Support for Memory Safety

A lot of work. The key is to understand the design trade-offs.

	Intel MPX (Memory Protection Extension)	**ARM MTE (Memory Tagging Extension)**
History	Announced in 2013, produced in 2015, now not supported anymore.	Introduced in ARM-8.5 in 2018. In 2019, Google announced that it is adopting Arm's MTE in Android. Apple will ship it soon.
Security		
Performance		
Compatibility		

Control-flow Integrity

- To maintain code pointer integrity
- Naïve idea:
  - Make pointer immutable (read-only)
  - Only work for global offset table and virtual function tables

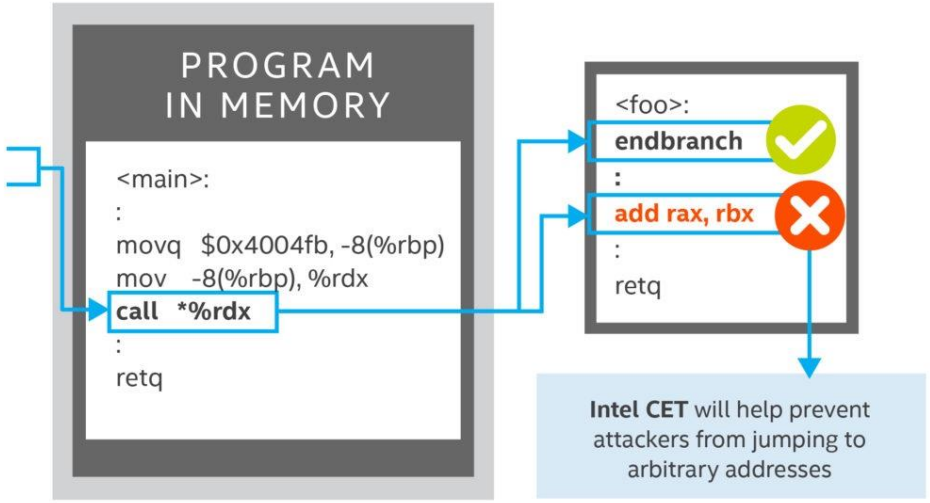
- How about other pointers?
  - Return address?
  - Programmer-defined function pointers
  - Change function pointers after changing vtable pointer

Intel® Control-Flow Enforcement Technology (Intel CET)

INTEL CET = INDIRECT BRANCH TRACKING (IBT) + SHADOW STACK (SS)

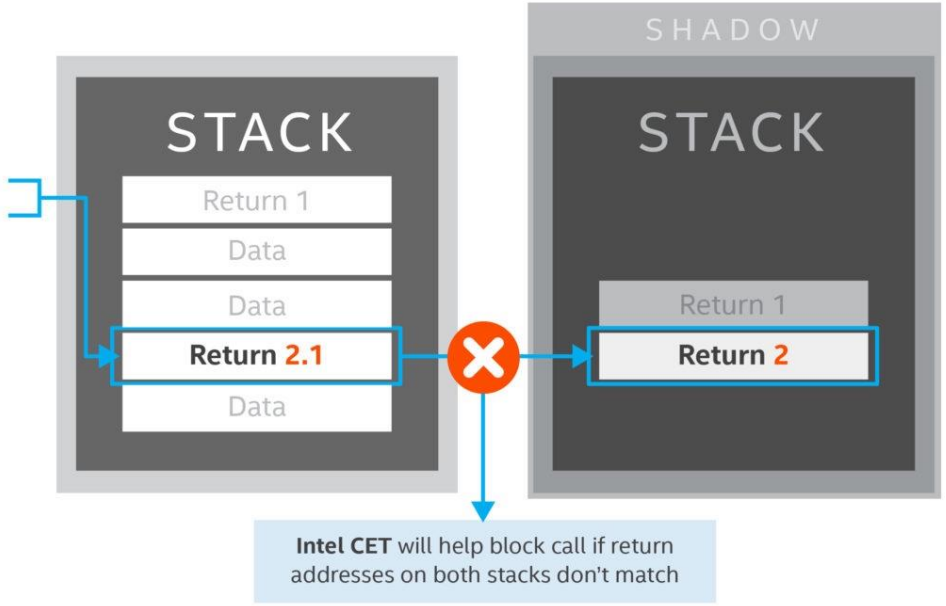
INDIRECT BRANCH TRACKING (IBT)

IBT delivers indirect branch protection to defend against jump/call oriented programming (JOP/COP) attack methods.



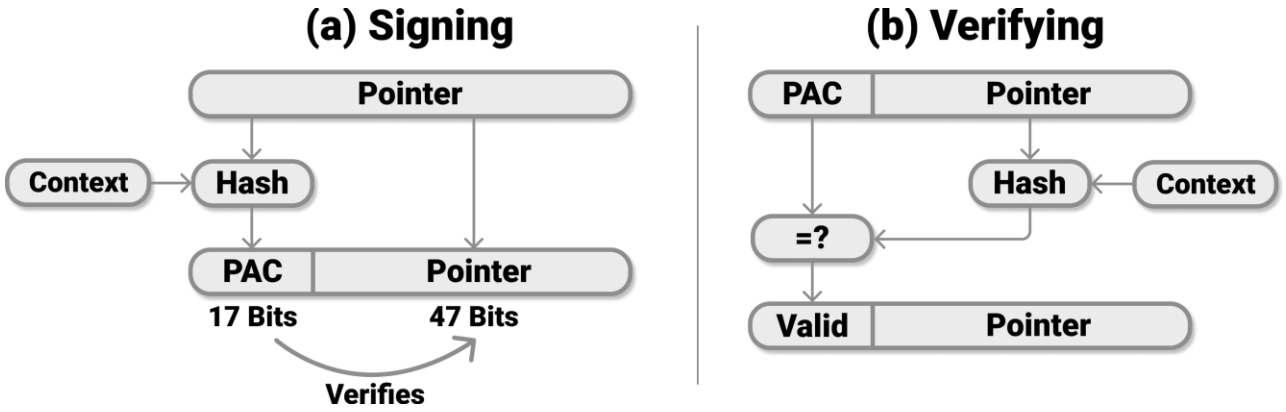
SHADOW STACK (SS)

SS delivers return address protection to defend against return-oriented programming (ROP) attack methods.



ARM PA (Pointer Authentication)

- Widely used in Apple processors
- Motivation:
  - 64-bit pointer, but 48-bit virtual address space
  - **Unused high bits**
  - Integrity check for pointers
- Hash: A tweakable message authentication code
  - `PAC = hash(key, context, pointer)`
  - Context can be `sp` or `vtable ptr`



Before function call

```
pacia lr, sp
sub   sp, sp, #0x40
str   lr, [sp, #0x30]
...
```

Before function return

```
ldr   lr, [sp, #0x30]
add   sp, sp, #0x40
autia lr, sp
ret
```

Summary

- Memory corruption problems: An eternal war
- Attack variations and mitigations
- Trade-off in hardware support

Enjoy the Spring Break!